

sorting_algo

April 2, 2025

```
[1]: # linear serach
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```
[2]: def linear_s(l1,item):  
    for i in range(len(l1)):  
        if l1[i] == item:  
            return i  
    return -1
```

```
[4]: l1 = [1,5,9,6,3,7]  
linear_s(l1,50)
```

```
[4]: -1
```

```
[8]: a = 0+4/2  
a
```

```
[8]: 2.0
```

```
[9]: int(a)
```

```
[9]: 2
```

```
[10]: # binary serach
```

```
[20]: def binary_s(arr,low,high,item):  
    print("low = ",low,"high = ",high,end=" ")  
    if low <= high:  
        mid = (low+high)//2  
        print("mid value",arr[mid])  
  
        if arr[mid] == item:  
            return mid  
        elif arr[mid] > item:  
            return binary_s(arr,low, mid-1,item)  
        else:  
            return binary_s(arr,mid+1,high,item)  
    else:  
        return -1
```

```
[21]: arr = [1,2,3,4,8,9,10,14,17]
      print(binary_s(arr,0,len(arr)-1,10))
```

```
low = 0 high = 8 mid value 8
low = 5 high = 8 mid value 10
6
```

```
[23]: # sorted or not
      def is_sorted(arr):
          sorted = True
          for i in range(len(arr) -1 ):
              if arr[i]>arr[i+1]:
                  sorted = False
          return sorted
```

```
[25]: arr = [1,2,3,4,5,4]
      is_sorted(arr)
```

```
[25]: False
```

```
[26]: def is_sorted(arr):
      for i in range(len(arr)-1):
          if arr[i] > arr[i+1]:
              return False
      return True
```

```
[28]: arr = [1,2,4,3]
      is_sorted(arr)
```

```
[28]: False
```

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[ ]:
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[ ]:
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